THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bulb rated 6 V is connected to a battery of three 1.5 V cells in series. The bulb will: 1

(a) not glow at all **(b)** glow dimly **(c)** glow normally **(d)** fuse

2. A student claims: 'Pure water conducts electricity well'. How would you test and correct this? 1

(a) add sugar to make it conduct
(b) test with a tester; add salt to water to make it conduct
(c) heat it
(d) water is a good conductor

3. A student sets up a tester with a cell, a bulb and two wires with free ends. The bulb glows when a steel spoon is placed across the ends. The spoon is: 1

(a) a fuse **(b)** a cell **(c)** a conductor **(d)** an insulator

4. A student connects five 1.5 V cells in series to a 6 V bulb. The bulb will: 1

(a) be dim **(b)** not glow
(c) glow very bright and may fuse **(d)** glow normally

5. A tester lights when connected across tap water but only dimly across distilled water. Reason: 1

(a) tap water has dissolved salts **(b)** distilled water is wet
(c) tap water is pure **(d)** distilled water has salts

6. In the same tester the bulb does not glow when a dry wooden pencil case is used. The case is: 1

(a) a filament **(b)** an insulator **(c)** a battery **(d)** a conductor

7. A chain of 10 decorative bulbs goes off when one bulb burns out. The bulbs are connected in: 1

(a) no circuit **(b)** open circuit **(c)** series **(d)** parallel

8. In the circuit tester, the bulb glows brightly with a copper wire and dimly with graphite. This shows: 1

(a) copper conducts better **(b)** graphite is an insulator
(c) both conduct equally **(d)** graphite conducts better

9. **Assertion (A):** A bulb glows when the switch in its circuit is closed. 1
Reason (R): A closed switch completes the circuit so that current can flow.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A fuse is connected in series with an electrical appliance. 1
Reason (R): Fuse wire has a very high melting point, so it never melts.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the filament of a bulb is made of a thin tungsten wire and not of copper. 2

12. Why are the pins of a plug made of brass or copper? 2

13. Why do electric sockets have covers for children's safety? 2

14. Why is the heating element usually a coil of nichrome and not of copper? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student connects 6 cells of 1.5 V each in series. Find the total voltage and say what happens to a 3 V bulb connected to it. 3

16. Why should heavy appliances be connected through thick wiring and not thin wiring? 3

17. Describe the safe way of fixing a bulb in a holder. 3

OR

Compare the heating of a thin and a thick wire carrying the same current and explain why a thin fuse wire is used. 3

18. Explain why a tungsten filament bulb glows white hot but an iron wire would not last. 3

OR

Explain what a series circuit is, giving one disadvantage. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ananya's torch has two cells of 1.5 V each placed one after another, a small bulb and a push switch. One day, even after pressing the switch, the bulb does not glow, though the cells are new. 5
(i) What is the total voltage of the two cells joined in series?
(ii) Name the component that opens or closes the circuit.
(iii) Name the thin wire inside the bulb that glows.
(iv) Give one possible reason for the bulb not glowing and explain it.

20. Plan an energy-saving programme for a home with five measures and say why each helps. 5

OR

Discuss the causes of electrical fires and ways to prevent them. 5

END OF QUESTION PAPER — Total: 40 Marks